

RESEARCH ARTICLE

Grazing Exclusion Modulates Preferential Flow Dynamics Through Root–Soil Architectural Reorganisation in Arid Shrublands

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ABSTRACT

Preferential flow (PF) critically regulates the water redistribution and ecological functions in arid ecosystems. This study investigated how grazed exclusion alters the root–soil architecture within PF zones and compared the structural differences between preferential and matrix flow (MF) zones. Focusing on *Caragana korshinskii* shrublands in Yanchi County, China, we employed CT scanning and 3D reconstruction to quantify the geometric patterns and topological characteristics of soil aggregates, macropores and root systems under contrasting management conditions (enclosed vs. grazed). Key findings revealed the following: (1) Enclosed natural grasslands exhibited maximum dye-stained areas (40.38%) and infiltration depths (271 mm); (2) Both grazed artificial and enclosed natural shrublands showed homogenised distributions of aggregates and macropores; (3) Enclosed practices significantly reduced aggregate number (47.2%–65.5%), macropore number (27.0%–30.9%) and root number (39.0%–64.5%) while promoting root thickening (10.7%–43.6% diameter increase, $p < 0.05$); (4) PF zones contained more aggregates (29.3%–219.1%) and macropore (35.3%–89.9%) than matrix zones ($p < 0.05$). The results show that moderate grazing can produce more aggregates and macropores, forming a root–soil structure that is more conducive to the PF process. These findings augment our understanding of rhizosphere interactions and sustainable land management practices.

1 | Introduction

Preferential flow (PF) is a pivotal hydrological process within unsaturated zones that has far-reaching impacts on global water cycles, ecological functions and environmental health. Characterised by spatial heterogeneity, rapid nonequilibrium infiltration, temporal fluctuation, universal occurrence and lateral redistribution (Zhang et al. 2024), PF plays dual roles in

ecosystems. It significantly enhances groundwater recharge in arid and semiarid regions worldwide, facilitating the survival and growth of drought-adapted vegetation (Zhang et al. 2022). However, the accelerated transport of contaminants through preferential pathways poses a substantial threat to riparian ecosystems and groundwater quality, jeopardising public health and ecological stability (Allaire et al. 2009; Hester and Fox 2020). The intricate interplay between PF and biogeochemical cycles

further underscores the urgent need for in-depth research in this field for effective ecosystem management and policymaking (Guo and Lin 2018; Jiang et al. 2020).

Given the critical role of the PF, understanding the factors that influence its behaviour is of utmost importance. One such fundamental factor is the soil structure. Soil structure, defined as the three-dimensional arrangement of particles and pores, serves as a fundamental framework that regulates root development and soil hydraulics (Sullivan et al. 2022). Compaction of the soil structure can impede root elongation and limit the water uptake and photosynthetic capacity of plants, which is a widespread concern in various agricultural and natural ecosystems (de Moraes et al. 2020; Mariotti et al. 2020). Notably, roots actively modify soil hydraulics via particle reorientation, pore network restructuring, aggregate reorganisation and macropore formation via decay (Lu et al. 2020). This reciprocal interaction between the roots and soil forms a dynamic root–soil continuum, which is essential for nutrient cycling, soil formation and overall ecosystem productivity (Ogilvie et al. 2021; Sofo et al. 2020). The advent of X-ray computed tomography has revolutionised the study of complex root–soil architectures, enabling nondestructive quantification and three-dimensional visualisation (Hou et al. 2022; Kuka and Joschko 2024).

As we delve deeper into the relationship between root–soil structure and PF, it becomes evident that land management practices can significantly alter this relationship. Grazing and enclosure are two prevalent land management practices employed globally to maintain and restore degraded ecosystems. In arid and semi-arid regions, which cover approximately one-third of the Earth's land surface, the effects of these management practices on soil structure, root development and PF dynamics are particularly significant. Enclosure measures are often implemented to promote vegetation restoration and soil conservation and can alter nitrogen cycling, biodiversity patterns and carbon sequestration over the long term (Bai et al. 2021; Chen et al. 2022; Yeneayehu et al. 2021). However, the effectiveness of enclosures in achieving sustainable ecosystem restoration may vary depending on the duration, intensity and specific ecological context (Xu et al. 2020). In contrast, moderate grazing can enhance carbon storage in arid ecosystems by redistributing assimilated carbon within the plant–soil system and preserving soil structural integrity through bioturbation and root turnover (Bai et al. 2021; Cao et al. 2023; Wang et al. 2023a).

Despite extensive research on the individual impacts of grazing and enclosure on various ecological processes (Worrall and Adamson 2008), how these management practices specifically modulate PF dynamics through root–soil architectural reorganisation remains unclear. Previous studies have primarily focused on the general effects of land management on soil properties and vegetation characteristics (Perri et al. 2017), with limited attention given to the specific mechanisms underlying root–soil structure interactions and PF across different ecological contexts. Moreover, the structural differences between PF and matrix flow (MF) zones under contrasting management regimes have not been fully explored. This lack of understanding hinders the development of comprehensive targeted strategies designed to optimise water redistribution and ecological functions in water-limited ecosystems.

This study aims to bridge these knowledge gaps by investigating the *Caragana korshinskii* shrublands in Yanchi County, China, a representative arid ecosystem facing the challenges of land degradation and water scarcity, using advanced CT scanning and 3D reconstruction techniques. We had three main objectives. First, we aimed to quantify the effects of the enclosure on the root–soil topology within the PF zones. Second, we compared the structural differences between the PF and MF zones. Finally, we sought to establish mechanistic links between the root–soil architecture and PF. Through these objectives, we expect to provide novel insights into the complex interactions between land management, root–soil structure and PF. These insights contribute to the development of sustainable land management practices and vegetation restoration strategies for water-limited ecosystems.

2 | Materials and Methods

2.1 | Study Area

The investigation was conducted in Yanchi County (37°57'23''–37°83'47'' N, 106°77'99''–107°50'70'' E), Ningxia Hui Autonomous Region, China, situated at 1600 m average elevation. The dominant soil types in the area are sierozem and aeolian sandy soils. Characterised by a temperate continental climate, the region exhibits a mean annual temperature of 8.1°C, with 150 frost-free days. Annual precipitation averages 280 mm, showing a strong seasonal concentration (> 80% of precipitation) from May to September, of which 60% falls during July to September. Evaporation rates reached 2710 mm/yr., exceeding precipitation by 9.7-fold. Westerly and north-westerly winds predominantly occurred during the spring months. The vegetation community consists of drought-adapted species assemblages, including *Lespedeza potaninii*, *Cynanchum komarovii*, *Astragalus melilotoides*, *Artemisia scoparia*, *Setaria viridis*, *Stipa breviflora*, *Sophora alopecuroides*, *Corispermum hyssopifolium* and *Heteropappus altaicus*, reflecting typical arid–semiarid ecosystem characteristics.

2.2 | Plot Selection and Sampling Methods

The field experiment was initiated in April 2022, and six plots were established based on vegetation types and grazing management regimes. All plots maintained consistent edaphic conditions (sandy grassland soil) and topographic features (1600 m elevation). Grazing exclusion had been implemented in the enclosed plots for over 20 years (Table 1).

At each plot, two representative sampling points containing both the PF zones and MF zones were selected. From each zone, soil cores were extracted from two depth intervals (0–10 cm and 10–20 cm) using custom-made steel cylinders (50 mm inner diameter × 65 mm height, wall thickness < 0.5 mm) designed to preserve the root–soil architecture. This sampling design yielded 48 cores (6 plots × 2 replicates × 2 flow zones × 2 depths) for subsequent analysis. Core extraction followed standardised protocols: (1) surface litter removal prior to sampler placement; (2) Vertical insertion using peripheral soil excavation to minimise structural disturbances; (3) reinforcement with steel plates at both cylinder

ends; (4) fresh-keeping film wrapping to prevent moisture loss and mechanical displacement. Immediately following collection, samples underwent X-ray microtomography using a nano-Voxel-5000 dual-source CT system (Sanying Precision Co. Ltd.). Three additional replicate samples per treatment were collected using conventional methods (aluminium boxes and sealable bags) for baseline soil characterisation (Table 2).

2.3 | Infiltration Dying Experiment

After a comprehensive literature review and field testing (Filipović et al. 2020; Wang et al. 2024), this experiment adopted the dye tracer method, using a custom-made aluminium alloy frame with a length, width, height and thickness of 120, 60, 30, and 0.5 cm respectively. For each mode, a replication structure was designed with two plot-level replicates and four sampling point replicates, resulting in eight independent experimental units. A sampling point with relatively flat terrain and uniform vegetation was selected as the test site, and the adjacent shrubland position of the test site was set as the soil dye tracer observation section. To ensure accurate results, the surface impurities of the soil were carefully removed without disturbing the humus layer. After cleaning, the aluminium alloy frame was inserted halfway into the soil along a 90°

angle and covered with plastic film to protect the sampling site from damage by animals or environmental factors. After 1 day, the plastic film was removed, and a bright blue dye solution was prepared based on the surface area of the dye sample and actual consumption. The spraying volume in the study area was determined based on the precipitation data from July to August. According to the calculations, the dye concentration in the study area was 4 g/L and required 11.5 L of solution, which was applied using a constant-flow pump for water infiltration on the sample site. After spraying, the plate was covered with plastic film, and the frame was removed. High-resolution digital images were taken of the 12 dye sections that were excavated vertically at the same interval.

2.4 | CT Scanning and Analysis of Root–Soil Structure Parameters

2.4.1 | CT Scanning and Image Processing

Forty-eight undisturbed soil samples were scanned using a NanoVoxel 5000 series dual-ray source CT system (Sanying Precision Instruments Co. Ltd., Tianjin, China; pixel detail resolution of 200 nm, spatial resolution of 200 nm and measurable sample size diameter×height not higher than

TABLE 1 | Basic information of plot.

Plot	Land management methods	Vegetation condition	Coordinate	Altitude/m
T1	Grazed	Artificial <i>Caragana korshinskii</i> + <i>Artemisia scoparia</i> + <i>Setaria viridis</i>	37°49'43" N, 107°30'08" E	1519
T2		Natural <i>Caragana korshinskii</i> + <i>Artemisia scoparia</i> + <i>Setaria viridis</i>	37°49'36" N, 107°30'37" E	1516
T3		<i>Artemisia scoparia</i> + <i>Setaria viridis</i>	37°49'42" N, 107°30'31" E	1517
T4	Enclosure	Artificial <i>Caragana korshinskii</i> + <i>Artemisia scoparia</i> + <i>Setaria viridis</i>	37°49'25" N, 107°31'11" E	1497
T5		Natural <i>Caragana korshinskii</i> + <i>Artemisia scoparia</i> + <i>Setaria viridis</i>	37°49'37" N, 107°30'38" E	1511
T6		<i>Artemisia scoparia</i> + <i>Setaria viridis</i>	37°49'31" N, 107°30'45" E	1518

TABLE 2 | Basic properties of plot soil. ± indicates the standard deviation.

Plot	Soil bulk density/ (g·m ⁻³)	Steady infiltration rate/(mm·min ⁻¹)	Soil moisture Content/%	Soil particle composition/%		
				< 0.05 mm	0.05~0.25 mm	> 0.25 mm
T1	1.37 ± 0.04	2.95 ± 0.20	7.89 ± 0.60	1.12	56.02	42.87
T2	1.46 ± 0.01	2.40 ± 0.11	8.32 ± 0.40	3.60	58.28	38.12
T3	1.42 ± 0.03	2.78 ± 0.23	4.56 ± 0.90	1.12	66.52	32.37
T4	1.31 ± 0.03	3.00 ± 0.07	10.54 ± 1.77	7.23	57.26	26.30
T5	1.30 ± 0.02	2.93 ± 0.13	5.76 ± 1.39	7.97	65.74	26.30
T6	1.31 ± 0.05	3.45 ± 0.23	7.35 ± 1.40	7.92	73.20	18.89

600 mm × 600 mm) (Figure 1a). After CT scanning, each undisturbed soil column yielded 2094 transverse slice images. First, ImageJ software (NIH, Bethesda, MD, USA) was used to preprocess all the cross-sectional images. Then, the image processing software Avizo 2019.1 (Thermo Fisher Scientific, Waltham, MA, USA) (Figure 1b) was used to reconstruct all the cross-sectional images of each soil column in three dimensions to obtain the geometric distribution images of each undisturbed soil sample. The three-dimensional reconstruction method is as follows: First, the binary image was imported and subjected to integration processing and threshold segmentation. Next, unit modification was conducted using the clipping function to ensure the reconstructed model closely resembles the real specimen. The volume-edit module was then used to capture the cylinder and generate a mask. Finally, a three-dimensional image was obtained with a voxel of 0.137 mm/pixel (X direction), 0.137 mm/pixel (Y direction) and 0.2 mm/pixel (Z direction).

2.4.2 | Analysis of the Root–Soil Structural Parameters

After reconstructing the geometric distribution image, the edge of the soil column exerts pressure on the soil in the opposite direction during sampling, causing soil structure changes at the edge. Therefore, it is necessary to remove errors caused by this edge effect. After observation and analysis, data within a 1 mm margin around the circumference and at the upper and lower ends of the soil column were removed using the ROI tool of the Avizo 2019.1 graphics software. This created a refined three-dimensional image dataset of the soil column with X and Y axes

of –24 to 24 mm and a Z axis of –65 to 0 mm, which was used for subsequent calculations.

Avizo 2019.1 was then used for detailed image analysis to characterise the distribution of soil roots, macropores and aggregates. Roots with a minimum diameter of 0.1 mm were identified, and all relevant parameters and spatial positions of root segments ≥ 0.1 mm in the 3D image were extracted. All pores (diameter > 0.1 mm) and aggregates (diameter > 0.25 mm) were extracted. The relevant parameters and their positions in the images were calculated. All extracted data were integrated into Microsoft Excel 2016 (Redmond, WA, USA) for further analysis. The topological structure indices of the root–soil are listed in Table 3.

2.5 | Image Processing and Data Analysis

ERDAS IMAGINE v9.2 was used to perform geometric correction on the dye-tracing image, and Photoshop CS6 was used for illumination correction, noise reduction and threshold processing. Finally, Image ProPlus v6.0 was used to perform morphological analysis, segmentation, counting, watershed processing and other functions. Independent, closed dye-stained clumps were divided, and the area, number and position were extracted. The dye-stained area ratio was then calculated. The formula for calculating the dyeing area ratio is given in Equation (1):

$$DC = \left(\frac{S_t}{S_n + S_t} \right) \times 100\% \quad (1)$$

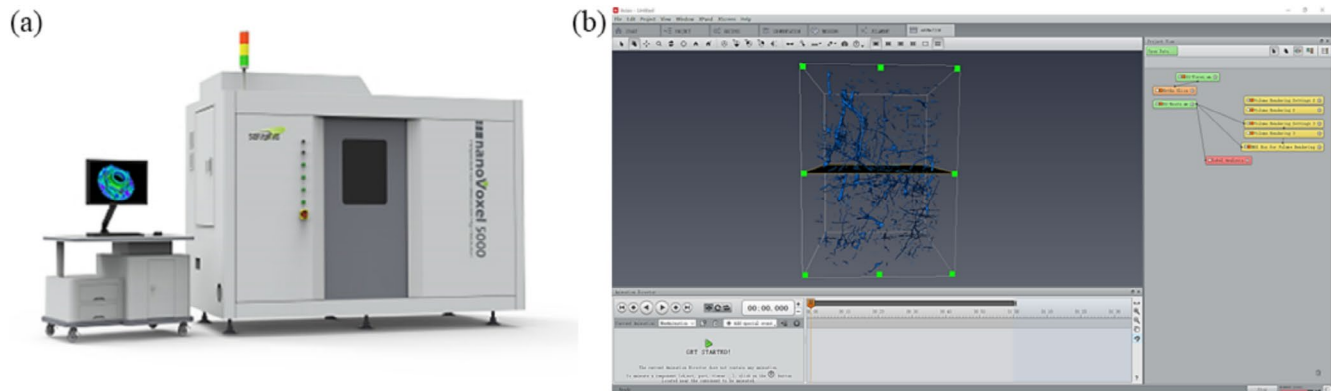


FIGURE 1 | CT scanning system (a) and image processing software (b).

TABLE 3 | Topological structure index of root–soil.

Topological structure index	Unit	Meaning
Number density	No.·mm ^{−3}	The number of root–soil structures (aggregates, macropores and roots) per cubic millimetre of soil column.
Length density	mm·mm ^{−3}	The total length of root–soil structure per cubic millimetre of soil column.
Surface area density	mm ² ·mm ^{−3}	The total surface area of the root–soil structure per cubic millimetre of soil column.
Volume density	mm ³ ·mm ^{−3}	The total volume of root–soil structure per cubic millimetre of soil column.
Average equivalent diameter	mm	The average equivalent diameter of root–soil structure.

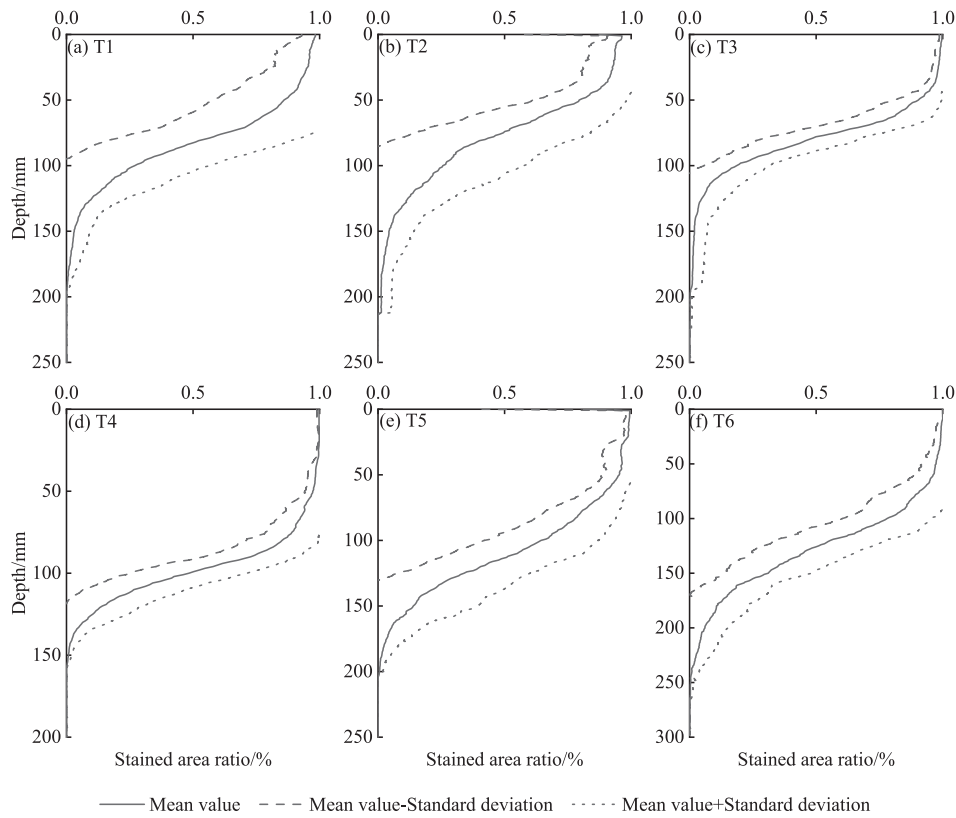


FIGURE 2 | Change of preferential flow area ratio with soil depth.

where DC is the ratio of the dying area (%), S_t is the total dying area of the soil profile (cm^2) and S_n is the total undyed area of the soil profile (cm^2).

2.6 | Statistical Analysis

Microsoft Excel 2016 was used to organise the experimental data. As the root–soil structure data did not conform to the homogeneity of variance, a nonparametric Kruskal–Wallis H test was conducted in R 4.0.3 for comparison and visualisation.

3 | Results

3.1 | Analysis of PF Characteristics

The percentage of dye coverage progressively declined with increasing soil depth across all treatments (Figure 2). In the 0–400 mm profile, maximum infiltration metrics clearly demonstrated the effects of land management. Dye coverage followed the order: enclosed grassland (40.38%) > natural *C. korshinskii* shrubland (39.86%) > enclosed artificial *C. korshinskii* shrubland (34.79%) > grazed artificial *C. korshinskii* shrubland (30.67%) > grazed natural *C. korshinskii* shrubland (30.62%) > grazed grassland (28.52%). The infiltration depths were as follows: enclosed grassland (271 mm) > enclosed natural *C. korshinskii* shrubland (226 mm) > grazed grassland (223 mm) > grazed natural *C. korshinskii* shrubland (213 mm) > enclosed artificial *C. korshinskii* shrubland (209 mm) > grazed artificial *C. korshinskii* shrubland (200 mm).

3.2 | The Difference of Aggregate Topological Structure in Different Plots

There were clear differences in the geometric distribution characteristics of soil aggregates in the different plots (Figure 3). In contrast, the distribution of soil aggregates in the grazed artificial, grazed natural and enclosed natural *C. korshinskii* shrublands was more uniform, with a high distribution density. In contrast, the distribution uniformity of soil aggregates in other plots was poor. In the enclosed artificial *C. korshinskii* shrublands and grasslands, the distribution of soil aggregates in the PF zone was more uniform and compact than in the MF zone.

Significant topological variations in the soil aggregates were observed between the PF and MF zones (Figure 4). In PF zones, artificial *C. korshinskii* shrublands exhibited dramatic reductions in aggregate densities (number: 65.5%, length: 60.0%, surface area: 55.0%, volume: 56.3%) alongside a 17.8% increase in mean equivalent diameter (all $p < 0.05$). Conversely, natural *C. korshinskii* shrublands exhibited enhanced connectivity under the enclosure, with a 21.3% higher length density and 9.6% larger aggregate size ($p < 0.05$). Meanwhile, natural grasslands displayed uniform declines across metrics (number: 47.2%, length: 45.2%, surface area: 37.9%, volume: 36.1% and diameter: 11.2%, $p < 0.05$).

Extreme values highlighted management contrasts: grazed *C. korshinskii* artificial shrublands attained the maximum number density (0.052 No. $\cdot \text{mm}^{-3}$) with minimal aggregate size (0.141 mm), while enclosed natural *C. korshinskii* shrublands achieved peak connectivity (length: 0.013 $\text{mm} \cdot \text{mm}^{-3}$;

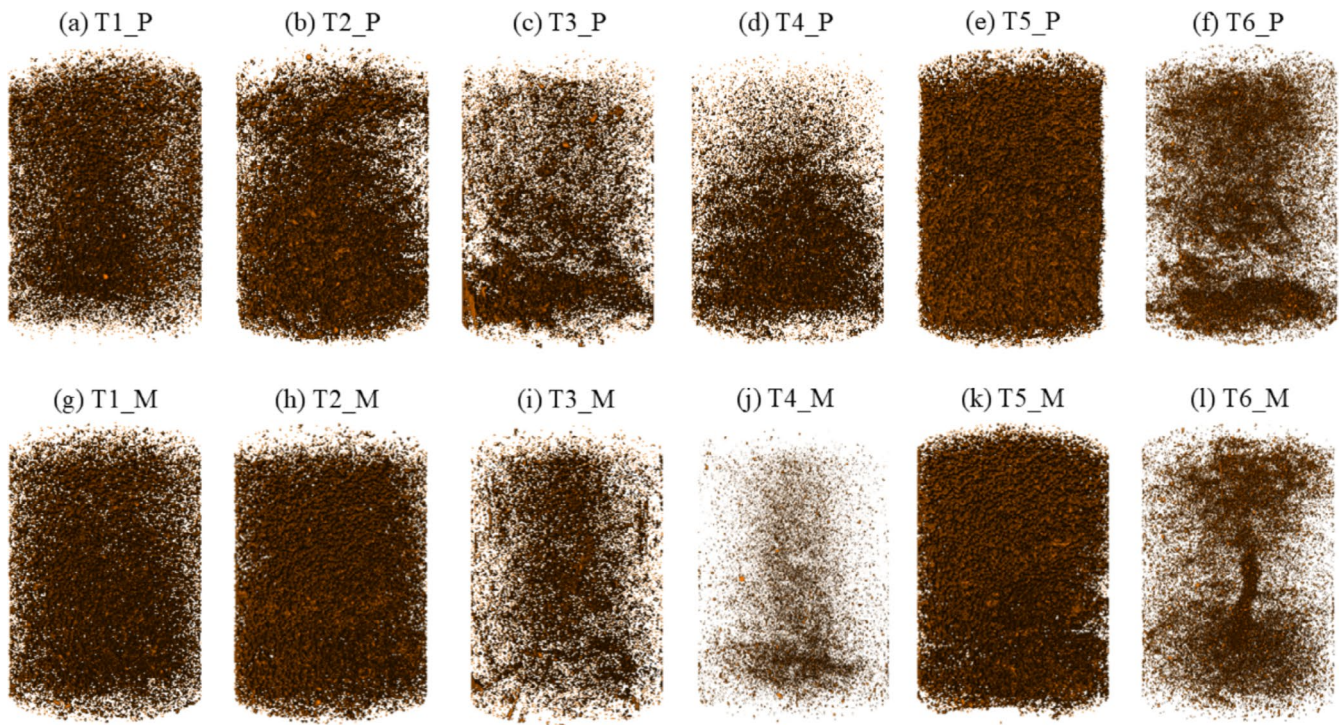


FIGURE 3 | Geometric distribution of soil aggregates in different plots (0–10 cm). P is the preferential flow zone; M is the matrix flow zone. The same below.

surface area: $0.009 \text{ mm}^2 \cdot \text{mm}^{-3}$) and largest diameter (0.178 mm). Enclosed natural grasslands recorded the lowest volume density ($0.00008 \text{ mm}^3 \cdot \text{mm}^{-3}$), whereas grazed natural *C. korshinskii* shrublands recorded the highest volume density ($0.00047 \text{ mm}^3 \cdot \text{mm}^{-3}$).

The PF zones demonstrated significantly enhanced aggregate metrics compared to the MF zones (Figure 4). The aggregate number density increased by 29.3%–219.1%, length density by 32.5%–101.9% and surface area density by 38.5%–50.2% (all $p < 0.05$). Conversely, the mean equivalent diameter exhibited a 10.6%–19.3% reduction in the PF zone ($p < 0.05$).

3.3 | The Difference of Macropore Topological Structure in Different Plots

It can be seen that there are differences in the geometric distribution characteristics of soil macropores in different plots (Figure 5). The distribution of soil macropores in grazed artificial *C. korshinskii* shrublands and enclosed natural *C. korshinskii* shrublands was more uniform, and the distribution density was high. In contrast, the distribution of soil macropores in other plots was poor, and the soil macropores were concentrated in some areas. In naturally grazed *C. korshinskii* shrublands, grazed grasslands and enclosed grasslands, the distribution of soil macropores in the PF zone was more uniform and compact than that in the MF zone.

Distinct macropore topological patterns emerged between the PF and MF zones across vegetation types (Figure 6). Enclosure induced differential effects. Artificial *C. korshinskii*

shrublands exhibited significant PF zone reductions in macropore density (number: 27.0%, length: 30.5%, surface area: 26.1%, volume: 21.6%, $p < 0.05$). Natural *C. korshinskii* shrublands showed a parallel decline (number: 30.9%, length: 27.2%), despite a 6.3% increase in the mean equivalent diameter ($p < 0.05$). No significant enclosure effects were observed in grasslands ($p > 0.05$).

Management extremes revealed striking contrasts. Maximum macropore density in grazed natural *C. korshinskii* shrublands was $0.035 \text{ No.} \cdot \text{mm}^{-3}$, $0.008 \text{ mm} \cdot \text{mm}^{-3}$ and $0.004 \text{ mm}^2 \cdot \text{mm}^{-3}$. The minimum macropore density in enclosed artificial *C. korshinskii* shrublands was $0.015 \text{ No.} \cdot \text{mm}^{-3}$, $0.004 \text{ mm} \cdot \text{mm}^{-3}$ and $0.002 \text{ mm}^2 \cdot \text{mm}^{-3}$. The highest macropore volume was observed in enclosed natural *C. korshinskii* shrublands (peak $0.00020 \text{ mm}^3 \cdot \text{mm}^{-3}$), while the lowest occurred in grazed grasslands (minimum $0.00012 \text{ mm}^3 \cdot \text{mm}^{-3}$). In terms of size, grazed artificial *C. korshinskii* shrublands recorded the largest macropores (0.137 mm), while enclosed grasslands exhibited the smallest (0.127 mm).

The PF zones exhibited contrasting macropore responses compared to the MF zones (Figure 6). Enclosed natural grasslands showed significant reductions in the PF zone macropore metrics: number density (24.5%), length density (26.7%), surface area density (31.7%) and volumetric density (30.3%) (all $p < 0.05$). In contrast, most of the other vegetation types demonstrated reversed patterns with substantial increases in PF zones: number density (35.3%–89.9%), length density (28.6%–56.2%), surface area density (21.3%–23.4%), volume density (11.4%–17.1%) (all $p < 0.05$). Concurrently, the mean equivalent diameter decreased by 2.6%–11.0% ($p < 0.05$).

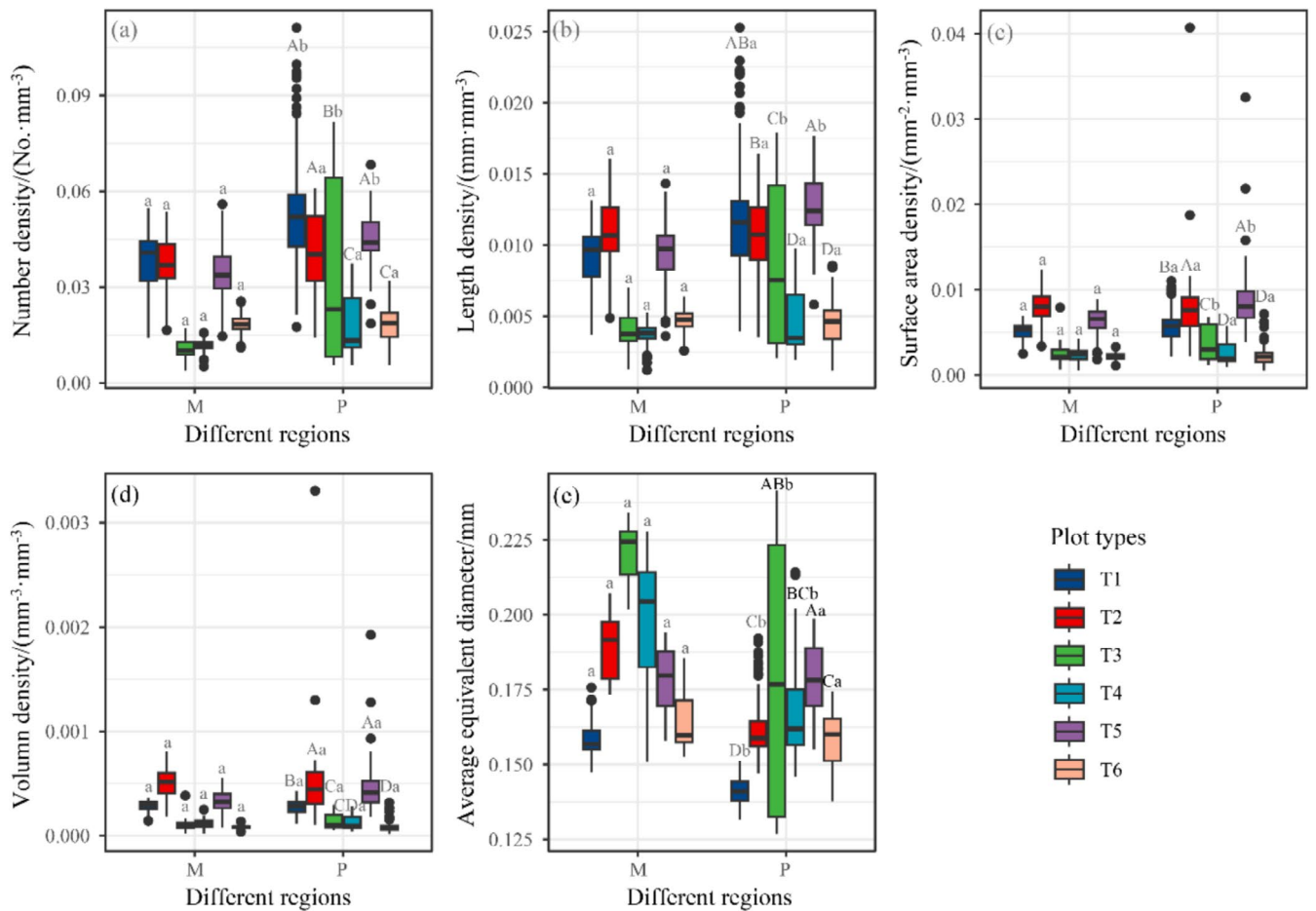


FIGURE 4 | Changes of soil aggregate topological structure in different plots. Different capital letters indicate that there are significant differences in soil topological structure in preferential flow zones of different plots ($p < 0.05$). Different lowercase letters indicate that there is a significant difference in soil topological structure between preferential flow and matrix flow zones ($p < 0.05$). The same below.

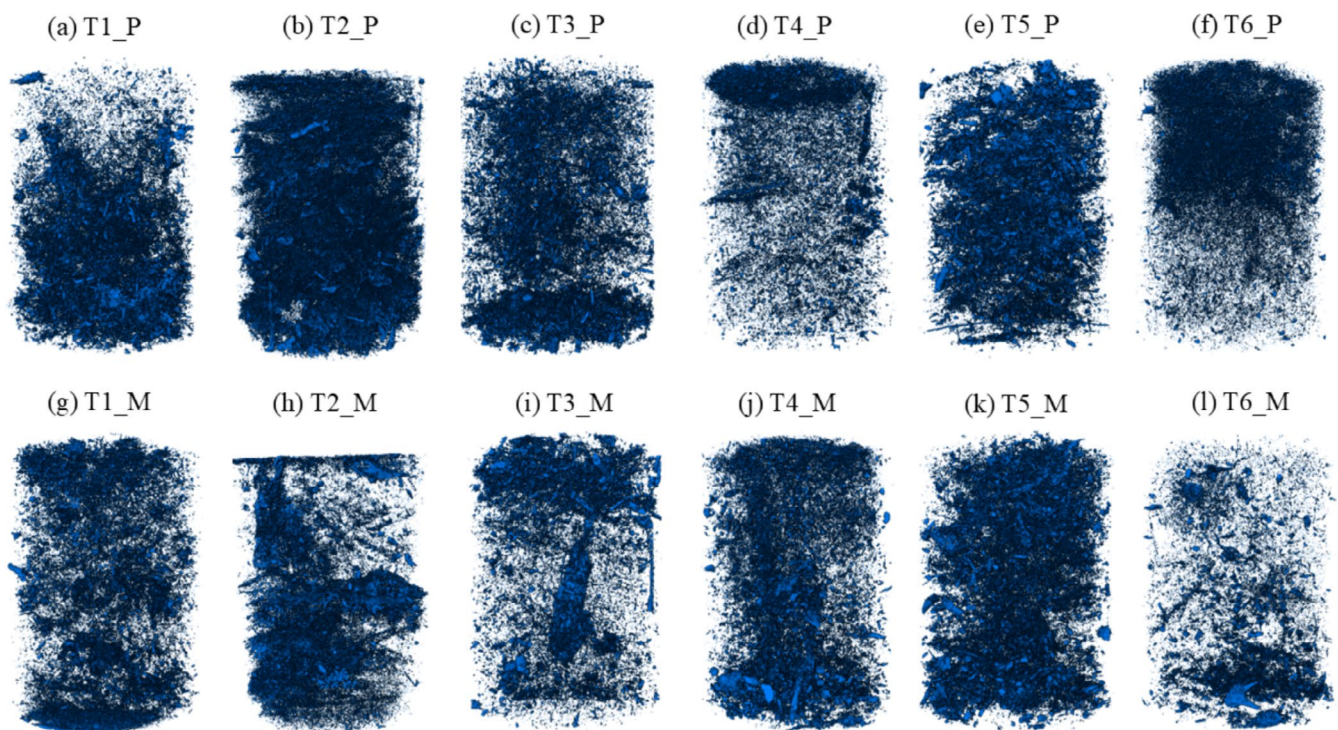


FIGURE 5 | Geometric distribution of soil macropores in different plots (0~10 cm).

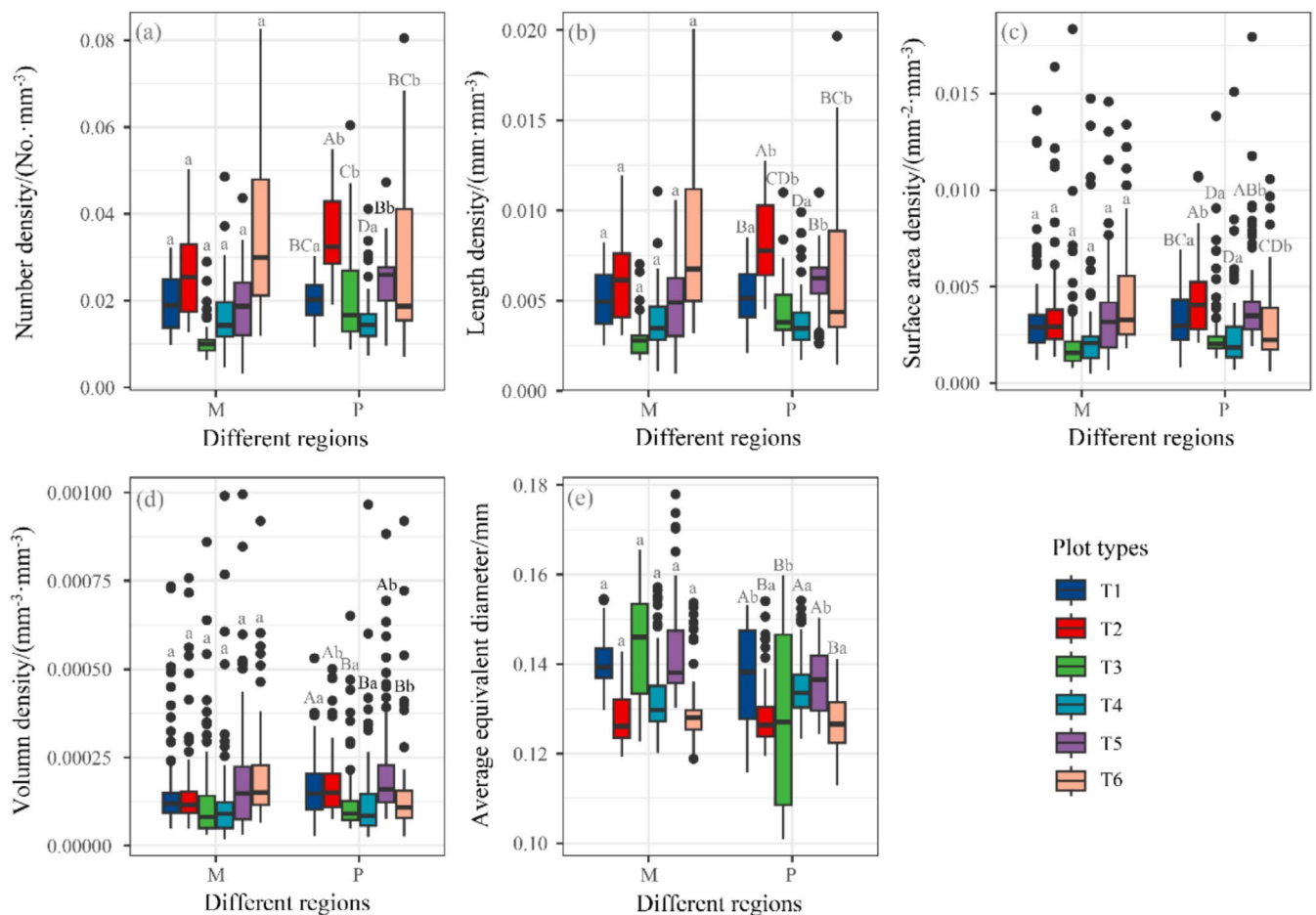


FIGURE 6 | Changes of soil macropore topological structure in different plots.

3.4 | The Difference of Root Topological Structure in Different Plots

The differences in the geometric distribution characteristics of soil roots varied across plots (Figure 7). In natural grasslands, root distribution was relatively uniform with high distribution density. In contrast, other plots exhibited lower uniformity, with roots concentrated in localised areas. The roots in *C. korshinskii* shrublands were thicker and more unevenly distributed than those in the natural grassland. In the artificially grazed *C. korshinskii* shrubland, the distribution of soil roots in the PF zone was more uniform and closer than that in the MF zone.

Significant variations in root topology were observed between the PF and MF zones across vegetation types (Figure 8). Enclosure differentially modulated root architecture. In artificial *C. korshinskii* shrublands, PF zones exhibited 39.0% lower root number density and 25.9% reduced root length density ($p < 0.05$), despite a 43.6% increase in mean equivalent diameter ($p < 0.05$). In the natural *C. korshinskii* shrublands, the volume density increased by 55.0%, with a 10.7% larger root diameter in the PF zones ($p < 0.05$). In natural grasslands, PF zones showed marked reductions in the following root metrics: number density (64.5%), length density (55.5%), surface area density (41.9%)

and volume density (31.4%) ($p < 0.05$), coupled with a 20.7% greater root diameter ($p < 0.05$).

Grazed natural grasslands exhibited the following maximum root density metrics: number density (0.00028 No.·mm⁻³), length density (0.00071 mm·mm⁻³), surface area density (0.00083 mm²·mm⁻³) and volume density (0.00006 mm³·mm⁻³)—values 1.97–2.8 times higher than minimums observed in enclosed natural grasslands (0.00010 No.·mm⁻³) and grazed natural *C. korshinskii* shrublands (0.00030 mm·mm⁻³, 0.00042 mm²·mm⁻³, 0.00003 mm³·mm⁻³). Notably, enclosed grasslands exhibited the largest mean root diameter (0.534 mm), in contrast to the smallest value observed in grazed artificial *C. korshinskii* shrublands (0.436 mm).

The PF zones exhibited distinct root architectural modifications compared to the MF zones (Figure 8). In naturally grazed *C. korshinskii* shrublands, the PF zones showed coordinated reductions in root metrics: number density (33.8%), length density (33.7%), surface area density (38.6%) and volume density (38.9%) (all $p < 0.05$). Grazed artificial *C. korshinskii* shrublands demonstrated a 15.6% reduction in root length density in the PF zone ($p < 0.05$). Conversely, enclosures enhanced root structural development by 8.9% and 10.5% in mean equivalent diameter for artificial *C. korshinskii* shrublands and natural grasslands respectively ($p < 0.05$).

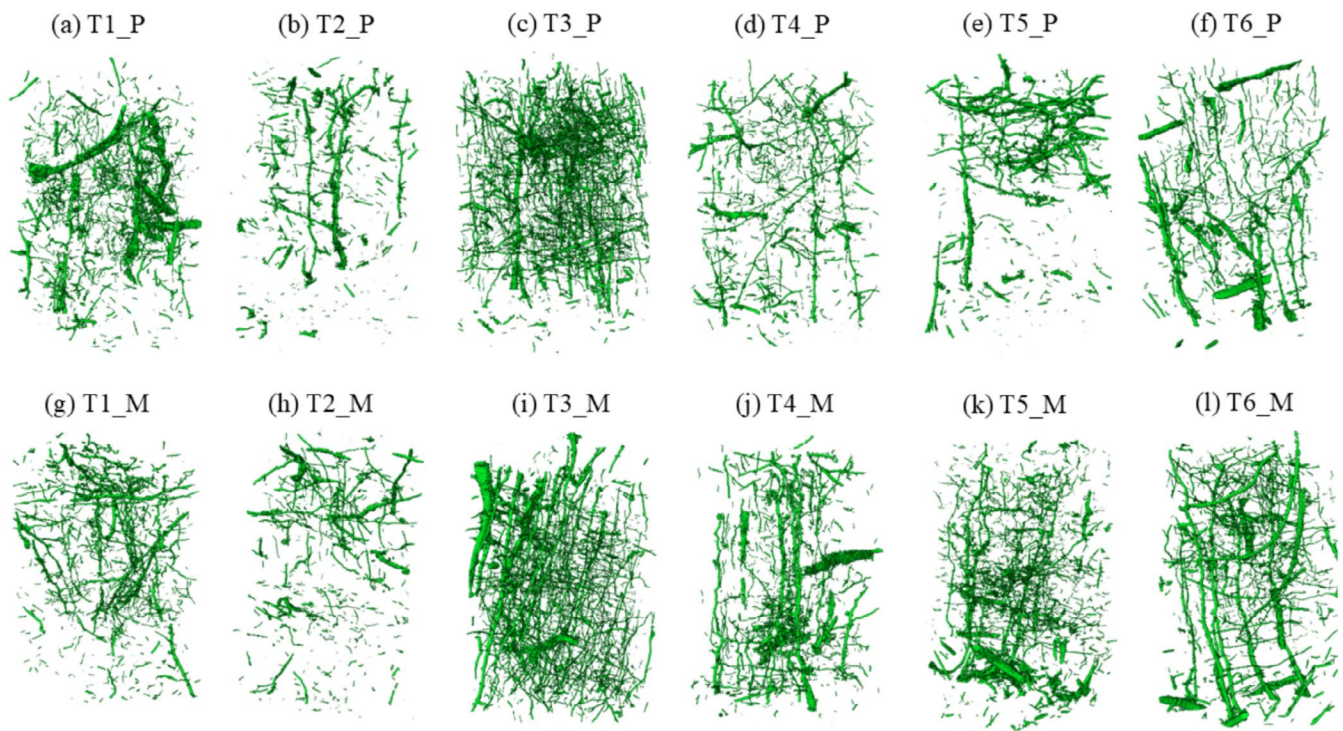


FIGURE 7 | Geometric distribution of soil roots in different plots (0~10 cm).

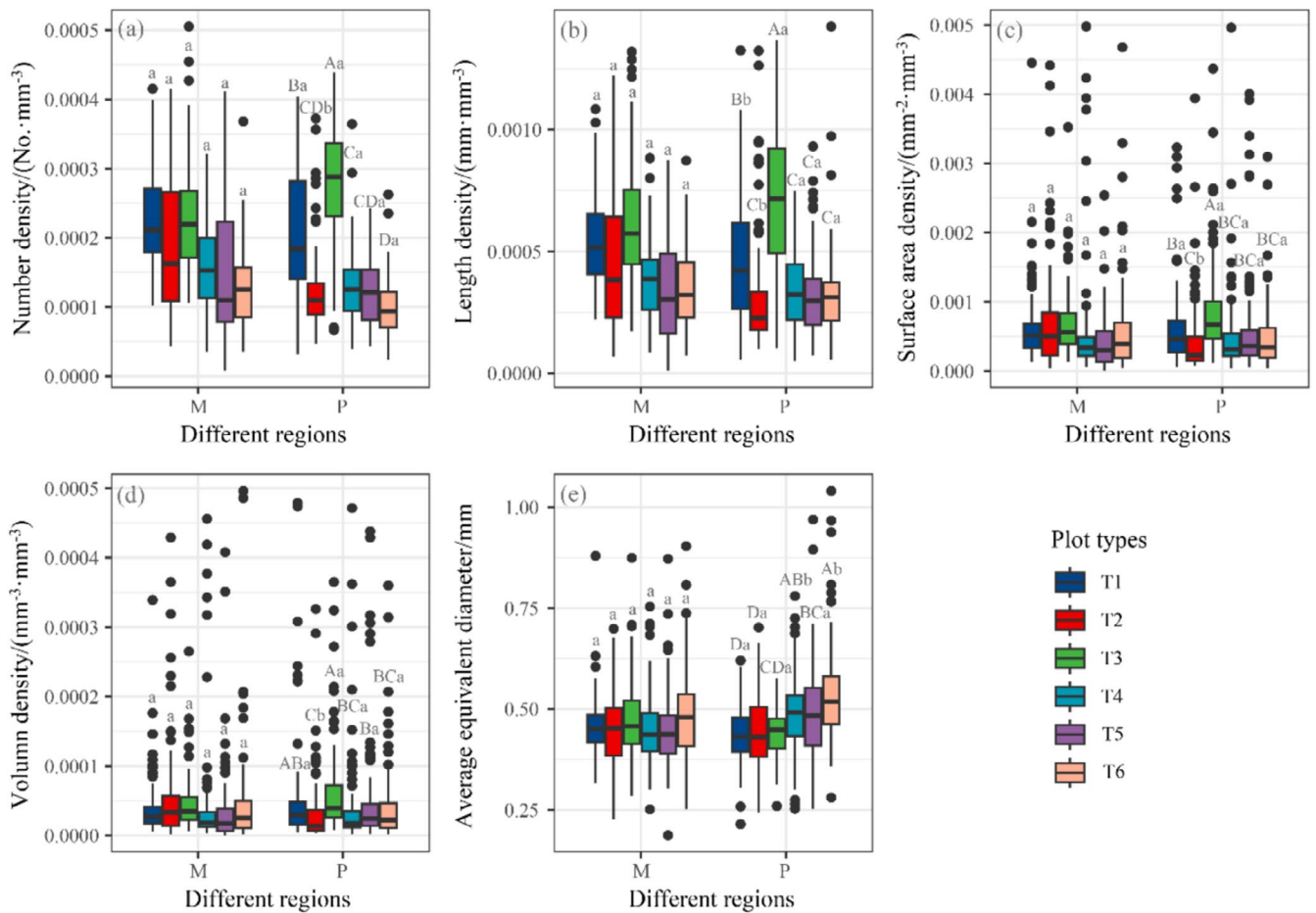


FIGURE 8 | Changes of soil root topological structure in different plots.

4 | Discussion

4.1 | Characteristics of PF in Different Plots

Our findings demonstrated superior dye coverage (40.38%) and infiltration depth (271 mm) in enclosed natural grasslands, aligned with established root architecture advantages present in diverse plant communities (Frank et al. 2010). The complex root networks of multispecies systems enhance macropore connectivity and facilitate deeper water penetration. Similarly, enclosed natural *C. korshinskii* shrublands exhibited greater dye depths than their grazed counterparts, likely because of improved aggregate stability after enclosure (Wei et al. 2022).

Notably, artificial *C. korshinskii* shrubland plantations exhibited reduced infiltration metrics compared with natural systems, regardless of the management regime. This impairment stems from initial planting disturbances that degrade the native soil architecture (Cui et al. 2023), particularly through the destruction of existing macropore networks. Nevertheless, artificial *C. korshinskii* shrublands increased ecosystem water retention capacity relative to degraded grasslands (Gou et al. 2023), suggesting trade-offs between rapid hydrological recovery and long-term structural development.

4.2 | Geometric Distribution Characteristics of Soil Structure

Our findings revealed superior spatial homogeneity of aggregates and macropores in grazed artificial *C. korshinskii* shrublands and enclosed natural *C. korshinskii* shrublands. Enclosure of natural shrublands ameliorates grazing-induced structural fragmentation, allowing the natural development of soil architecture through enhanced plant diversity (Angassa and Oba 2010).

Conversely, artificial shrubland plantations exhibited compensatory structural development under grazing. The synergistic effects of grazing and planting disturbances enhanced macropore connectivity, despite the initial soil disruption. Natural grasslands demonstrate a higher root distribution uniformity than shrublands (Wang et al. 2023b), which explains their superior PF metrics.

Notably, in the enclosed artificial *C. korshinskii* shrubland and grassland, the distribution of soil aggregates in the PF zone was more uniform and compact than that in the MF zone. In naturally grazed *C. korshinskii* shrublands, grazed grasslands and enclosed grasslands, the distribution of soil macropores in the PF zone was more uniform and compact than that in the MF zone. Such architectural optimisation facilitates rapid water transmission through coordinated pore-aggregate interactions (Vervoort et al. 1999), where aggregate stability mediates PF initiation and macropores govern flow sustainability.

4.3 | Characteristics of Soil Aggregates in PF Zone

Grazing exclusion differentially modulated soil aggregate dynamics in PF zones across vegetation types. Enclosed practices significantly reduced aggregate number density (65.5% in artificial *C. korshinskii* shrublands and 47.2% in natural grasslands, $p < 0.05$)

and volume density (56.3% and 36.1%, respectively, $p < 0.05$). This is likely due to excessive litter accumulation altering nutrient cycling and aggregate stability (Dong et al. 2022). Vegetation-specific responses were evident in size distribution patterns. PF zones of enclosed *C. korshinskii* shrublands exhibited 9.6%–17.8% larger mean aggregate diameters ($p < 0.05$), whereas enclosed grasslands showed an 11.2% diameter reduction ($p < 0.05$).

These divergent trends stem from root architectural adaptations. Grassland dense fine-root networks promote smaller microaggregate formation via rhizodeposition, whereas the woody tap-roots of *C. korshinskii* shrubs stabilise larger microaggregates through root–soil entanglement (Okolo et al. 2020). Enclosure amplifies these intrinsic processes, promoting root development in undisturbed *C. korshinskii* shrublands and the formation of larger microaggregates. In contrast, fine root proliferation in grasslands enhances the generation of smaller microaggregates.

Notably, the PF zones contained 29.3%–219.1% more aggregates than the MF zones ($p < 0.05$), with 10.6%–19.3% smaller mean diameters ($p < 0.05$). This size-selective pattern aligns with the finding that aggregate size is one of the determinants of the rapid infiltration pathway (Manns et al. 2024), establishing an inverse abundance–size relationship characteristic of rapid infiltration systems.

4.4 | Characteristics of Soil Macropores in PF Zone

Grazing exclusion induced contrasting effects on the macropore architecture in *C. korshinskii* shrublands. Enclosed practices significantly reduced the macropore number density in the PF zones by 27.0% (artificial *C. korshinskii* shrublands) and 30.9% (natural *C. korshinskii* shrublands), and the volume density decreased by 21.6% in the artificial *C. korshinskii* shrublands ($p < 0.05$). This suppression contrasts with the capacity of grazing to maintain soil structural integrity through shrub-mediated pore preservation, in which woody roots stabilise macropores while enhancing infiltration rates (Eldridge et al. 2015; Raube et al., Rauber et al. 2021).

Notably, enclosed natural *C. korshinskii* shrublands exhibited a 6.3% increase in the PF zone macropore diameter ($p < 0.05$), reflecting postenclosure root maturation and organic matter accumulation (Hu et al. 2018). Comparative analysis revealed that the PF zones contained 35.3%–89.9% more macropores and 11.4%–17.1% greater volume density than the MF zones ($p < 0.05$), indicating spatially extensive macropore networks (Nieber and Sidle 2010). Paradoxically, these PF zones showed 2.6%–11.0% smaller mean macropore diameters ($p < 0.05$), suggesting that PF pathways become dominated by finer, more numerous conduits, a pattern consistent with the observed microaggregate predominance in these zones. Such structural adaptations likely optimise subsurface water transmission in arid shrubland ecosystems while maintaining soil stability.

4.5 | Characteristics of Soil Roots in PF Zone

Grazing exclusion affected the root architecture of *C. korshinskii* in shrublands and natural grasslands. Enclosure practices

significantly reduced root number density in the PF zones by 39.0% (artificial *C. korshinskii* shrublands) and 64.5% (natural grasslands) ($p < 0.05$), reflecting suppressed root proliferation in the absence of grazing-induced soil disturbance (Bonin et al. 2013). Conversely, enclosed systems exhibited 10.7%–43.6% larger mean root diameters ($p < 0.05$), indicating a shift toward resource-conservative root-thickening strategies in response to grazing exclusion (Hu et al. 2018).

Management extremes revealed critical trade-offs. Grazed natural grasslands maintained the maximum root number density ($0.00028 \text{ No. mm}^{-3}$), suggesting rapid root turnover to compensate for grazed losses. In contrast, enclosed natural grasslands showed minimal root density ($0.00010 \text{ No. mm}^{-3}$), highlighting the strong suppression of new root growth after enclosure. Compared to the MF zones, the PF zones in the grazed natural *C. korshinskii* shrublands exhibited significant reductions in root abundance (number density: 33.8%, volume density: 38.9%, $p < 0.05$). These patterns suggest that root decay may enhance PF processes by reducing obstruction in the infiltration pathways (Noguchi et al. 1999).

Notably, enclosed *C. korshinskii* shrublands developed coarser roots in the PF zones, which facilitated macropore formation through root–soil interactions (Bodner et al. 2014). Thicker roots create stable pore networks, whereas finer roots dominate rapid infiltration pathways. This size-dependent root architecture indicates vegetation-specific strategies that balance soil exploration and hydrological functions in water-limited ecosystems.

4.6 | How Enclosure Affects the Root–Soil Structure and Thus Affects the PF Process

Grazing exclusion modulates PF processes through the structural reorganisation of the root–soil system (Figure 9). While enclosure practices reduced key structural metrics (aggregate number: 47.2%–65.5%, macropore number: 27.0%–30.9%, root

number: 39.0%–64.5%, $p < 0.05$), grazed maintained critical soil architecture by sustaining bioturbation and root turnover—processes essential for preserving macropore networks and aggregate stability (Cao et al. 2023). This suggests that moderate grazing disturbance acts as an ecosystem engineer, preventing structural homogenisation in arid soils.

The tripartite system of aggregates, macropores and roots exhibits coupled dynamics (Stumpf et al. 2016; Zhao et al. 2020). For aggregate–macropore feedback, aggregates stabilise pore walls while allowing selective particle transport. Regarding root–pore synergy, coarse roots create preferential pathways that enhance water transmission. Meanwhile, for disturbance adaptation, grazing-induced root pruning stimulates fine root proliferation, maintaining hydraulic connectivity.

Notably, they maintained 35.3%–89.9% higher macropore numbers and 29.3%–219.1% greater aggregate numbers in the PF zones than in the MF zones ($p < 0.05$). These structural advantages explain the enhanced infiltration capacity under grazed conditions. Our results align with the ecological threshold theory that when the belowground structural complexity exceeds critical levels, the preferential flow becomes the dominant hydrological process.

5 | Conclusions

Grazing exclusion significantly alters preferential flow dynamics through root–soil architectural reorganisation in arid lands. Enclosed practices reduced aggregate number (47.2%–65.5%), macropore number (27.0%–30.9%) and root number (39.0%–64.5%) while promoting root thickening (10.7%–43.6% diameter increase, $p < 0.05$). In contrast, moderate grazing maintained optimal architectural heterogeneity through bioturbation and root turnover, sustaining higher macropore and aggregate numbers, which enhanced the infiltration capacity. The self-reinforcing interaction between aggregates, macropores and roots creates size-selective pathways that favour rapid water

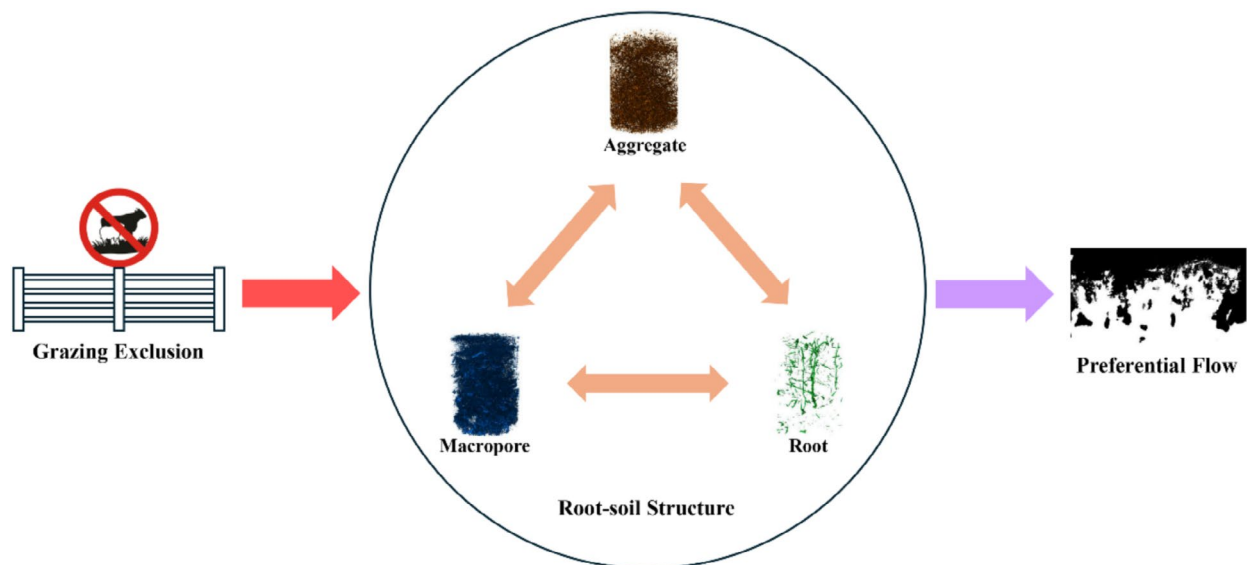


FIGURE 9 | Grazing exclusion modulates preferential flow dynamics through root–soil architectural reorganisation.

transport, challenging conventional enclosure-based restoration paradigms. Based on our findings, implementing adaptive grazing regimes to preserve the structural thresholds essential for water redistribution in water-limited ecosystems, while prioritising species-specific root architecture studies to optimise hydrological functions is necessary.

5.1 | Novelty and International Appeal Statement

Computed tomography scanning was used to construct the topological parameters of root–soil structure. The influence of enclosure on root–soil structure and the structural characteristics of root–soil in the preferential flow region were studied. We clarify how enclosure affects preferential flow processes by affecting root–soil structure.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

References

- Allaire, S. E., S. Roulier, and A. J. Cessna. 2009. "Quantifying Preferential Flow in Soils: A Review of Different Techniques." *Journal of Hydrology* 378, no. 1–2: 179–204. <https://doi.org/10.1016/j.jhydrol.2009.08.013>.
- Angassa, A., and G. Oba. 2010. "Effects of Grazing Pressure, Age of Enclosures and Seasonality on Bush Cover Dynamics and Vegetation Composition in Southern Ethiopia." *Journal of Arid Environments* 74, no. 1: 111–120. <https://doi.org/10.1016/j.jaridenv.2009.07.015>.
- Bai, X., X. Yang, S. Zhang, and S. An. 2021. "Newly Assimilated Carbon Allocation in Grassland Communities Under Different Grazing Enclosure Times." *Biology and Fertility of Soils* 57: 563–574. <https://doi.org/10.1007/s00374-021-01549-1>.
- Bodner, G., D. Leitner, and H.-P. Kaul. 2014. "Coarse and Fine Root Plants Affect Pore Size Distributions Differently." *Plant and Soil* 380, no. 1–2: 133–151.
- Bonin, C., J. Flores, R. Lal, and B. Tracy. 2013. "Root Characteristics of Perennial Warm-Season Grasslands Managed for Grazing and Biomass Production." *Agronomy* 3, no. 3: 508–523. <https://doi.org/10.3390/agronomy3030508>.
- Cao, J., Y. Chen, Y. Jiang, J. Chen, and J. Wu. 2023. "Management of Re-Established Artificial Grasslands via Grazing or Fencing: Effects on Plant and Soil Properties." *Grassland Research* 2, no. 1: 69–83. <https://doi.org/10.1002/glr2.12039>.
- Chen, Q., Y. Shang, R. Zhu, Q. Bao, and S. Lin. 2022. "Long-Term Enclosure at Heavy Grazing Grassland Affects Soil Nitrification via Ammonia-Oxidizing Bacteria in Inner Mongolia." *Scientific Reports* 12: 21464. <https://doi.org/10.1038/s41598-022-25367-z>.

- Cui, R., C. Wang, F. Cheng, et al. 2023. "Effects of Successive Planting of Eucalyptus on Soil Physicochemical Properties 1-3 Generations After Converting Masson Pine Forests Into Eucalyptus Plantations." *Polish Journal of Environmental Studies* 32, no. 5: 4503–4514.
- de Moraes, M. T., H. Debiase, J. C. Franchini, et al. 2020. "Soil Compaction Impacts Soybean Root Growth in an Oxisol From Subtropical Brazil." *Soil and Tillage Research* 200: 104611. <https://doi.org/10.1016/j.still.2020.104611>.
- Dong, L., Y. Zheng, V. Martinsen, C. Liang, and J. Mulder. 2022. "Effect of Grazing Exclusion and Rotational Grazing on Soil Aggregate Stability in Typical Grasslands in Inner Mongolia, China." *Frontiers in Environmental Science* 10: 844151. <https://doi.org/10.3389/fenvs.2022.844151>.
- Eldridge, D. J., G. Beecham, and J. B. Grace. 2015. "Do Shrubs Reduce the Adverse Effects of Grazing on Soil Properties?" *Ecohydrology* 8, no. 8: 1503–1513. <https://doi.org/10.1002/eco.1600>.
- Filipović, V., J. Defterdarović, J. Šimůnek, et al. 2020. "Estimation of Vineyard Soil Structure and Preferential Flow Using Dye Tracer, X-Ray Tomography, and Numerical Simulations." *Geoderma* 380: 114699. <https://doi.org/10.1016/j.geoderma.2020.114699>.
- Frank, D. A., A. W. Pontes, E. M. Maine, et al. 2010. "Grassland Root Communities: Species Distributions and How They Are Linked to Aboveground Abundance." *Ecology* 91, no. 11: 3201–3209. <https://doi.org/10.1890/09-1831.1>.
- Gou, Q., M. Gao, and G. Wang. 2023. "Multi-Functional Characteristics of Artificial Forests of *Caragana Korshinskii* Kom With Different Plantation Ages in the Hilly and Sandy Area of Northwest Shanxi, China." *Land Degradation & Development* 34, no. 14: 4195–4207. <https://doi.org/10.1002/ldr.4769>.
- Guo, L., and H. Lin. 2018. "Addressing Two Bottlenecks to Advance the Understanding of Preferential Flow in Soils." *Advances in Agronomy* 147: 61–117. <https://doi.org/10.1016/bs.agron.2017.10.002>. Elsevier.
- Hester, E. T., and G. A. Fox. 2020. "Preferential Flow in Riparian Groundwater: Gateways for Watershed Solute Transport and Implications for Water Quality Management." *Water Resources Research* 56, no. 12: e2020WR028186. <https://doi.org/10.1029/2020WR028186>.
- Hou, L. H., W. Gao, Z. H. Weng, et al. 2022. "Use of X-Ray Tomography for Examining Root Architecture in Soils." *Geoderma* 405: 115405. <https://doi.org/10.1016/j.geoderma.2021.115405>.
- Hu, X., Y. Lyu, Y. Liu, et al. 2018. "Exclosure on CT-Measured Soil Macropore Characteristics in the Inner Mongolia Grassland of Northern China." *Journal of Soils and Sediments* 18: 718–726. <https://doi.org/10.1007/s11368-017-1828-8>.
- Jiang, X. J., S. Zakari, J. Wu, et al. 2020. "Can Complementary Preferential Flow and Non-Preferential Flow Domains Contribute to Soil Water Supply for Rubber Plantation?" *Forest Ecology and Management* 461: 117948. <https://doi.org/10.1016/j.foreco.2020.117948>.
- Kuka, K., and M. Joschko. 2024. "Grassland Management Intensity Determines Root Development, Soil Structure, and Their Interrelationship: Results of a Regional Study of Leptosols in the Swabian Alps." *Grassland Research* 3: 171–186. <https://doi.org/10.1002/glr2.12077>.
- Lu, J., Q. Zhang, A. D. Werner, Y. Li, S. Jiang, and Z. Tan. 2020. "Root-Induced Changes of Soil Hydraulic Properties-A Review." *Journal of Hydrology* 589: 125203. <https://doi.org/10.1016/j.jhydrol.2020.125203>.
- Manns, H. R., Y. Jiang, and G. Parkin. 2024. "Soil Pores in Preferential Flow Terminology and Permeability Equations." *Vadose Zone Journal* 23, no. 5: e20365. <https://doi.org/10.1002/vzj2.20365>.
- Mariotti, B., Y. Hoshika, M. Cambi, et al. 2020. "Vehicle-Induced Compaction of Forest Soil Affects Plant Morphological and Physiological Attributes: A Meta-Analysis." *Forest Ecology and Management* 462: 118004. <https://doi.org/10.1016/j.foreco.2020.118004>.

- Nieber, J. L., and R. C. Sidle. 2010. "How Do Disconnected Macropores in Slope Soils Facilitate Preferential Flow?" *Hydrological Processes* 24, no. 12: 1582–1594. <https://doi.org/10.1002/hyp.7633>.
- Noguchi, S., Y. Tsuboyama, R. C. Sidle, and I. Hosoda. 1999. "Morphological Characteristics of Macropores and the Distribution of Preferential Flow Pathways in a Forested Slope Segment." *Soil Science Society of America Journal* 63, no. 5: 1413–1423. <https://doi.org/10.2136/sssaj1999.6351413x>.
- Ogilvie, C. M., W. Ashiq, H. B. Vasava, and A. Biswas. 2021. "Quantifying Root-Soil Interactions in Cover Crop Systems: A Review." *Agriculture* 11, no. 3: 218. <https://doi.org/10.3390/agriculture11030218>.
- Okolo, C. C., G. Gebresamuel, A. Zenebe, M. Haile, and P. N. Eze. 2020. "Accumulation of Organic Carbon in Various Soil Aggregate Sizes Under Different Land Use Systems in a Semi-Arid Environment." *Agriculture, Ecosystems & Environment* 297: 106924. <https://doi.org/10.1016/j.agee.2020.106924>.
- Perri, S., F. Viola, L. V. Noto, and A. Molini. 2017. "Salinity and Periodic Inundation Controls on the Soil-Plant-Atmosphere Continuum of Gray Mangroves." *Hydrological Processes* 31, no. 6: 1271–1282. <https://doi.org/10.1002/hyp.11095>.
- Rauber, L. R., L. Sequinatto, D. R. Kaiser, et al. 2021. "Soil Physical Properties in a Natural Highland Grassland in Southern Brazil Subjected to a Range of Grazing Heights." *Agriculture, Ecosystems & Environment* 319: 107515. <https://doi.org/10.1016/j.agee.2021.107515>.
- Sofo, A., H. S. Elshafie, and I. Camele. 2020. "Structural and Functional Organization of the Root System: A Comparative Study on Five Plant Species." *Plants* 9, no. 10: 1338. <https://doi.org/10.3390/plants9101338>.
- Stumpf, L., E. A. Pauletto, and L. F. S. Pinto. 2016. "Soil Aggregation and Root Growth of Perennial Grasses in a Constructed Clay Minesoil." *Soil and Tillage Research* 161: 71–78. <https://doi.org/10.1016/j.still.2016.03.005>.
- Sullivan, P. L., S. A. Billings, D. Hirmas, et al. 2022. "Embracing the Dynamic Nature of Soil Structure: A Paradigm Illuminating the Role of Life in Critical Zones of the Anthropocene." *Earth-Science Reviews* 225: 103873. <https://doi.org/10.1016/j.earscirev.2021.103873>.
- Vervoort, R. W., D. E. Radcliffe, and L. T. West. 1999. "Soil Structure Development and Preferential Solute Flow." *Water Resources Research* 35, no. 4: 913–928. <https://doi.org/10.1029/98WR02289>.
- Wang, D., J. Niu, Z. Dai, et al. 2024. "Forest Restoration Effects on Soil Preferential Flow in the Paleo-Periglacial Eastern Liaoning Mountainous Regions, China." *Journal of Cleaner Production* 467: 142974. <https://doi.org/10.1016/j.jclepro.2024.142974>.
- Wang, H., Y. Li, Y. He, et al. 2023a. "Grazing Exclusion Facilitates More Rapid Ecosystem Carbon Sequestration of Degraded Grasslands in Humid Than in Arid Regions." *Agriculture, Ecosystems & Environment* 353: 108553. <https://doi.org/10.1016/j.agee.2023.108553>.
- Wang, P., X. Su, Z. Zhou, N. Wang, and B. Zhu. 2023b. "Differential Effects of Soil Texture and Root Traits on the Spatial Variability of Soil Infiltrability Under Natural Revegetation in the Loess Plateau of China." *Catena* 220: 106693. <https://doi.org/10.1016/j.catena.2022.106693>.
- Wei, Y., M. Liu, J. Wang, X. Dang, and Y. Han. 2022. "The Effects of Vegetation Communities on Soil Organic Carbon Stock in an Enclosed Desert-Steppe Region of Northern China." *Soil Science and Plant Nutrition* 68, no. 2: 284–294. <https://doi.org/10.1080/00380768.2022.2045847>.
- Worrall, F., and J. K. Adamson. 2008. "The Effect of Burning and Sheep Grazing on Soil Water Composition in a Blanket Bog: Evidence for Soil Structural Changes?" *Hydrological Processes* 22, no. 14: 2531–2541. <https://doi.org/10.1002/hyp.6848>.
- Xu, L., Y. Nie, B. Chen, et al. 2020. "Effects of Fence Enclosure on Vegetation Community Characteristics and Productivity of a Degraded Temperate Meadow Steppe in Northern China." *Applied Sciences* 10, no. 8: 2952. <https://doi.org/10.3390/app10082952>.
- Yeneayehu, F., Y. You, X. Xu, and Y. Wang. 2021. "Effects of Grazing Enclosures on Species Diversity, Phenology, Biomass, and Carrying Capacity in Borana Rangeland, Southern Ethiopia." *Frontiers in Ecology and Evolution* 8: 623627. <https://doi.org/10.3389/fevo.2020.623627>.
- Zhang, J., T. Liu, L. Duan, et al. 2022. "Processes of Preferential Flow in a Eurasian Steppe Under Different Scenarios." *Journal of Hydrology* 612: 128166. <https://doi.org/10.1016/j.jhydrol.2022.128166>.
- Zhang, W., L. Wang, J. Chen, and Y. Zhang. 2024. "Preferential Flow in Soils: Review of Role in Soil Carbon Dynamics, Assessment of Characteristics, and Performance in Ecosystems." *Eurasian Soil Science* 57: 814–825. <https://doi.org/10.1134/S1064229323602548>.
- Zhao, Y., X. Hu, and X. Li. 2020. "Analysis of the Intra-Aggregate Pore Structures in Three Soil Types Using X-Ray Computed Tomography." *Catena* 193: 104622. <https://doi.org/10.1016/j.catena.2020.104622>.